

Surname	Centre Number	Candidate Number
First name(s)		0



GCSE
3445UA0-1



THURSDAY, 13 JUNE 2024 – MORNING

APPLIED SCIENCE (Double Award)
UNIT 1: Energy, Resources and the Environment
HIGHER TIER
1 hour 30 minutes

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use a pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **5** is a quality of extended response (QER) question where your writing skills will be assessed.
You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.
A Periodic Table is printed on page 20.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	19	
2.	9	
3.	13	
4.	16	
5.	6	
6.	12	
Total	75	

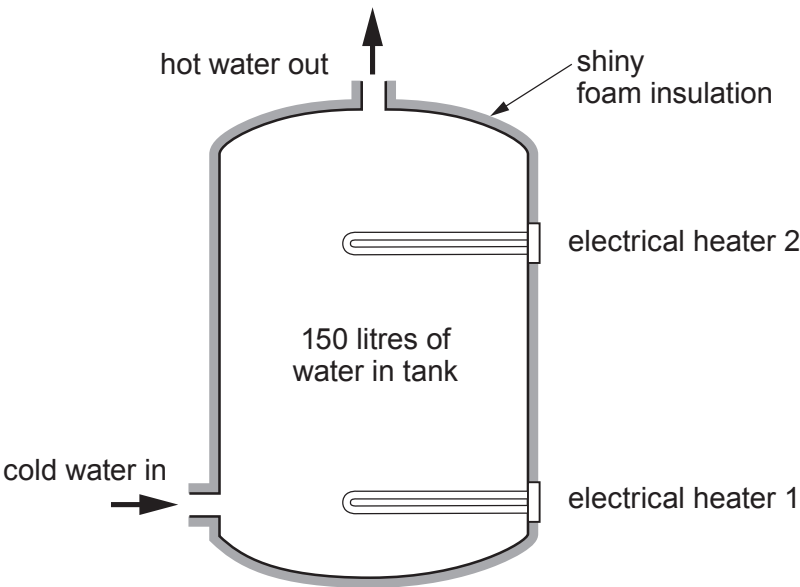


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Answer **all** questions.

1. Paul lives in an apartment in central London. His electricity bill is £80 a month. Paul wants to replace his old electric boiler with a new one. A local plumbing company has suggested using a new boiler, the Redstar boiler, which has two electrical heaters. Paul will be able to use either heater to provide hot water after the boiler is installed.

Information about the Redstar boiler



Savings

Cost of boiler	£2500.00
Typical savings	£15.00 per month

Technical information on the two heaters in the Redstar boiler

Information	heater 1	heater 2
maximum volume of water that is heated by heater (litres)	150	50
time to heat this volume of water (hours)	3	0.5
total power supplied (kW)	2	4
power used to heat the water (kW)	1.6	3.6
efficiency (%)	80	
cost to heat the water (p)		44

(a) Use the information given about the boiler to answer the following questions.

- (i) Calculate the expected payback time for the boiler in years.

[3]

payback time = years



(ii) Use the equation:

$$\% \text{ efficiency} = \frac{\text{power usefully transferred}}{\text{total power supplied}} \times 100$$

to calculate the efficiency of **heater 2**.

[2]

% efficiency =

(iii) Use the equations:

$$\text{units used (kWh)} = \text{power (kW)} \times \text{time (h)}$$

$$\text{total cost} = \text{cost of one unit} \times \text{units used}$$

to calculate the cost of the electricity to heat 150 litres of water using **heater 1**.

Cost per unit of electricity = 22p

[3]

cost =

(iv) Paul has estimated that he uses 40 litres of hot water per day.
He thinks that **heater 1** would be the best one to use every day.

Explain whether you agree with Paul.

[2]

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- (b) The water in London is known to be hard.

Explain how hard water could have affected Paul's old boiler.

[2]

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- (c) Heat energy can be transferred by conduction, convection and radiation.

- (i) Explain how the shiny foam insulation used around the boiler helps to reduce heat loss. [2]

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- (ii) Explain how having the foam insulation layer around the boiler benefits the environment. [2]

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- (iii) Paul thinks that **heater 2** will not heat all the water in the boiler.

Use the diagram of the boiler to explain whether you agree.

[3]

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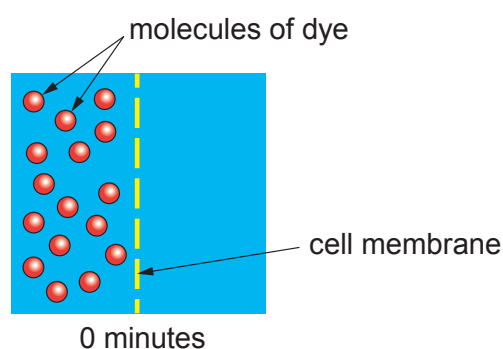
2. A student, Lisa, is studying how substances are transported into and out of cells.

(a) Oxygen enters the cell by diffusion. Define the process of diffusion. [1]

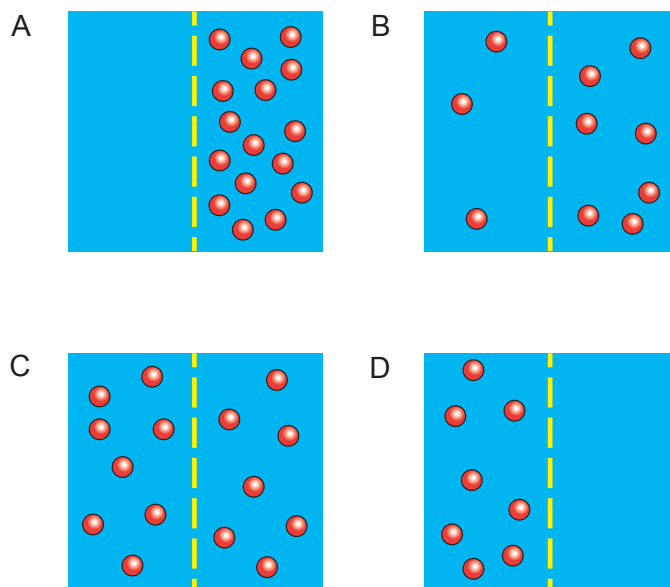
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(b) In her schoolbook, Lisa has diagrams of a model to show diffusion of dye into a cell. The dye takes 30 minutes to reach equilibrium.



Lisa thinks the diagram that would show the correct arrangement of dye molecules she would expect to see after 30 minutes is diagram A.



Explain whether you agree with Lisa.

[3]

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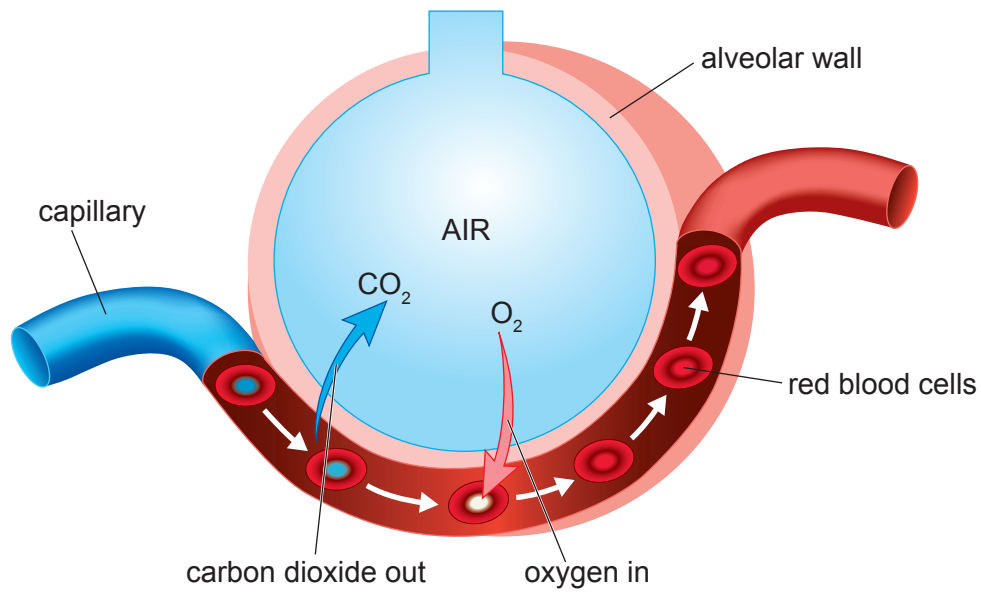
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- (c) Oxygen and carbon dioxide diffuse into and out of blood surrounding the alveoli.



Use the diagram to explain how diffusion is always maintained.

[3]

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- (d) Describe **two** differences between diffusion and active transport.

[2]

1.

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2.

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3. Welsh Water is responsible for supplying and maintaining safe water supplies in Wales. It monitors the composition of our drinking water.

The table below shows the composition of stream water supplying a reservoir in Wales. The samples were taken from three different streams.

Component	Concentration (units)		
	stream 1	stream 2	stream 3
sodium ions (Na ⁺)	5.0	5.0	5.0
potassium ions (K ⁺)	6.3	6.2	6.2
magnesium ions (Mg ²⁺)	15.3	50.4	60.5
calcium ions (Ca ²⁺)	50.4	50.1	101.1
chloride ions (Cl ⁻)	30.1	30.0	30.2
hydrogen carbonate ions (HCO ₃ ⁻)	10.1	50.1	40.0
dissolved sulfur dioxide	9.9	12.3	15.0
dissolved carbon dioxide	0.6	0.9	1.6
pH	5.7	5.5	5.4

Use the information in the table and your knowledge to answer the following questions.

(a) Explain how human activity has affected the acidity of the streams. [3]

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(b) A technician thinks the water in stream 1 is the softest and in stream 3 is the hardest. Explain whether you agree. [3]

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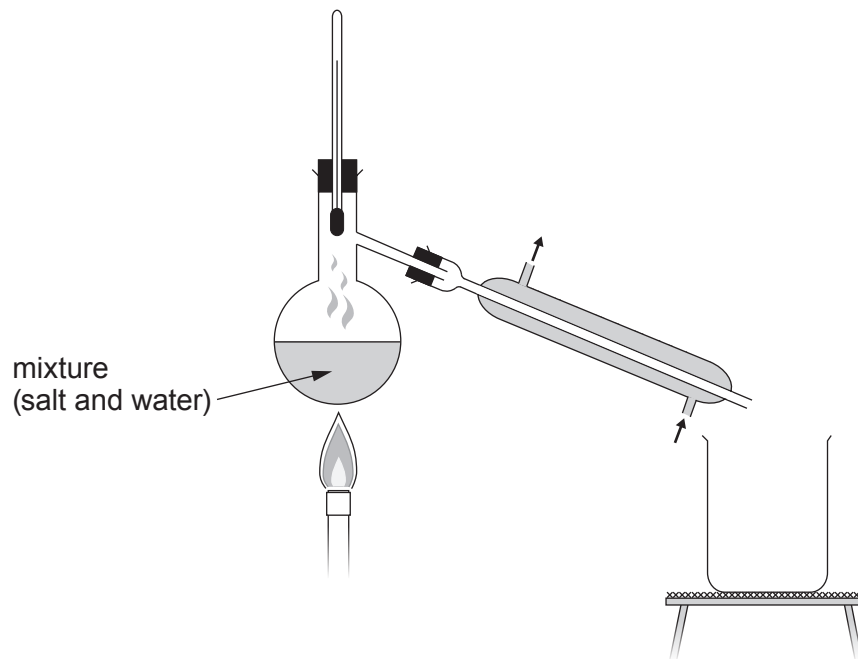
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- (c) Seawater can be desalinated using the apparatus below.



- (i) Describe how pure water is extracted using the apparatus above.

[4]

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- (ii) Desalination can be used to increase the water supply for the public in parts of the world where water is scarce.

Describe the factors a government must consider when investing in industrial-sized desalination plants.

[3]

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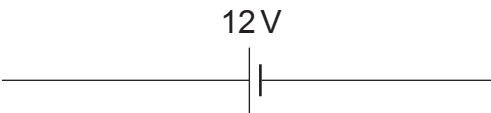
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4. Jonathon is investigating how the resistance of a thermistor varies with temperature.

(a) Complete the circuit diagram used to measure the resistance of a thermistor. [3]



(b) Jonathon’s results are shown in the table below.

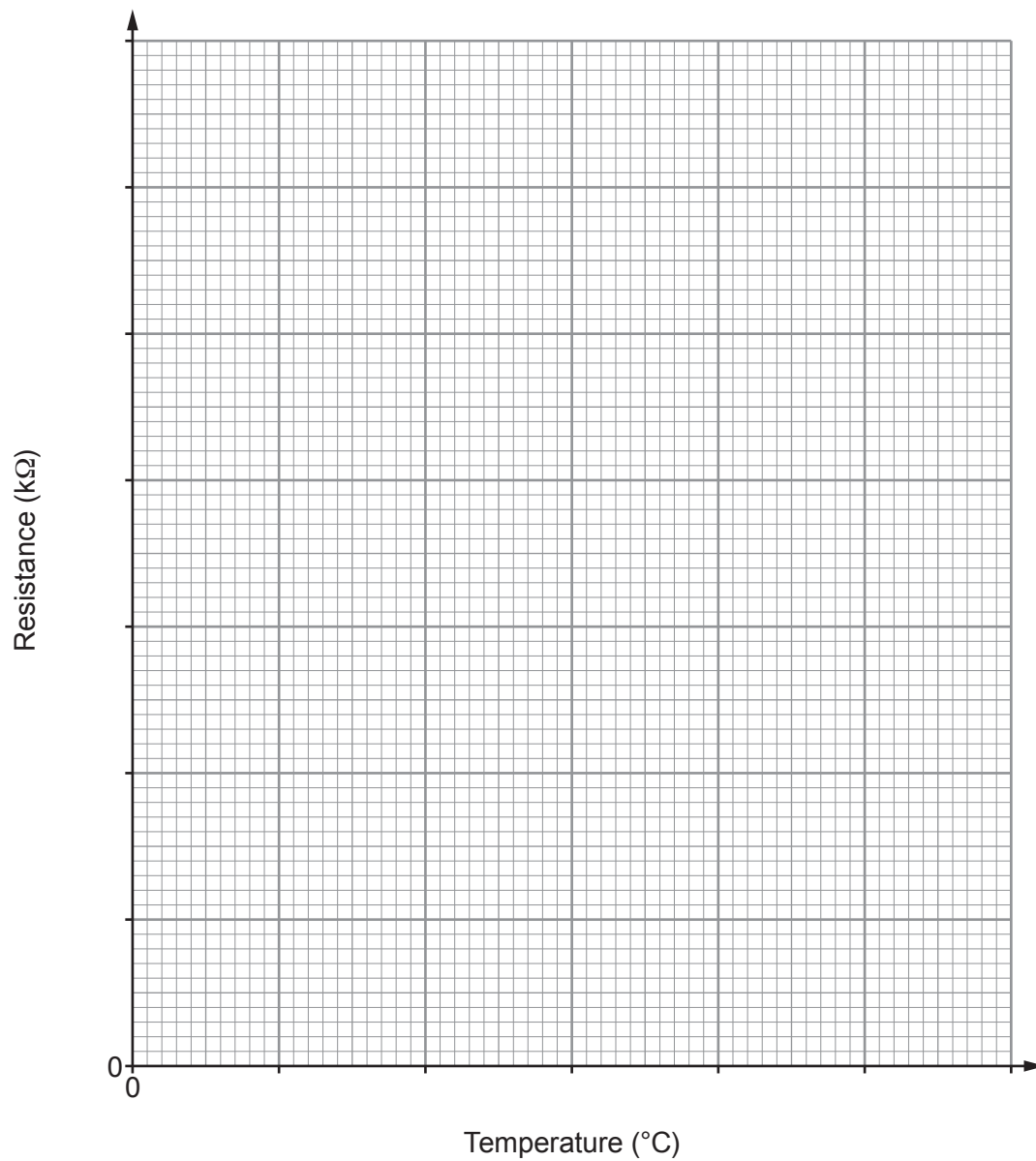
Temperature (°C)	Current (mA)	Voltage (V)	Resistance (kΩ)
20	1.01	12.0	11.9
40	2.21	12.0	5.4
60	4.78	12.0	2.5
80	8.51	12.0	1.4
100	19.95	12.0	0.6



(i) Plot the data on the grid and draw a suitable line.

[3]

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(ii) Describe the relationship shown by the graph.

[2]

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(iii) Use the graph and a suitable equation from the list below to answer each of the following questions.

resistance (Ω) = $\frac{\text{voltage (V)}}{\text{current (A)}}$

power = voltage $\times$ current

power = current² $\times$ resistance

I. Calculate the current through the thermistor at 30 °C. [4]

current =mA

II. Calculate the power developed in the thermistor at 30 °C. [2]

power =W

(c) Thermistors are found in refrigerators to control the temperature.

Describe how the experiment needs to be modified to test if this thermistor would be suitable for a refrigerator. [2]

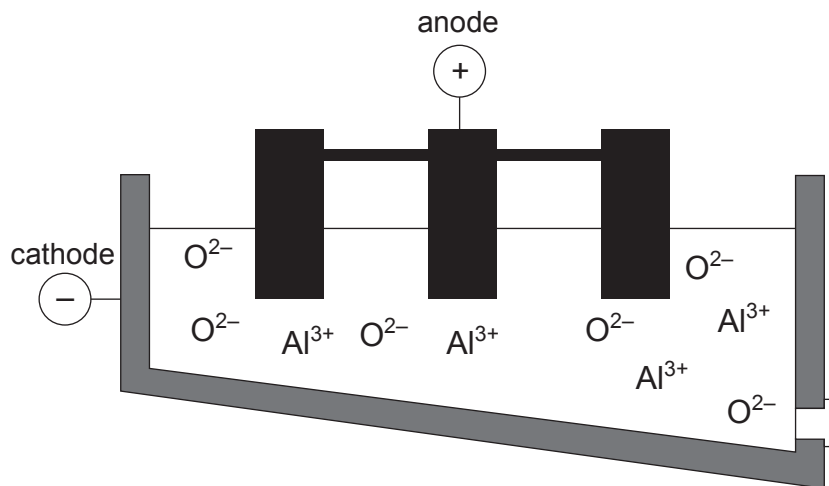
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5. Aluminium is extracted from its ore by electrolysis.



Use the diagram above to describe how aluminium is extracted from its ore.

[6 QER]



6. The table below shows the mean carbon footprint for 1 kWh of electricity generated in different ways.

Type of power station	Carbon footprint (gCO ₂ eq/kWh)
coal	921
oil	750
gas	500
solar	100
biomass	52
nuclear	30
hydroelectric	27
wind	30

Use the information in the table and your knowledge to answer the following questions.

(a) (i) Define the term carbon footprint. [2]

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(ii) A typical 500 MW **coal**-fired power station produces 3.5 billion kWh of electricity per year.
Calculate the carbon footprint of this power station in gCO₂eq/kWh per year.
Give your answer to 2 significant figures.

1 billion kWh = 1 × 10⁹ kWh [4]

carbon footprint = gCO₂eq/kWh



- (iii) The carbon footprint of a nuclear power station is close to zero whilst generating electricity.

Explain why the table shows a carbon footprint of $30 \text{ gCO}_2\text{eq/kWh}$ for power generated by nuclear power. [2]

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- (iv) A biomass power station burns wood chips as its energy source.
State why a biomass generator is considered as being 'carbon neutral'. [1]

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- (b) The Antarctic ice sheet traps methane. It is estimated that the mass of trapped methane is about $2.8 \times 10^{17} \text{ kg}$ which has a carbon dioxide equivalent of $7 \times 10^{18} \text{ kgCO}_2\text{eq}$.

- (i) Explain what is meant by the term global warming potential. [1]

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- (ii) Use the equation:

carbon dioxide equivalent = mass of a gas $\times$ global warming potential of the gas
to calculate the global warming potential of methane. [2]

global warming potential =

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THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1																		4 He Helium 2																			
7 Li Lithium 3			9 Be Beryllium 4															19 F Fluorine 9		20 Ne Neon 10																	
23 Na Sodium 11			24 Mg Magnesium 12															35.5 Cl Chlorine 17		40 Ar Argon 18																	
39 K Potassium 19			40 Ca Calcium 20			45 Sc Scandium 21		48 Ti Titanium 22		51 V Vanadium 23		52 Cr Chromium 24		55 Mn Manganese 25		56 Fe Iron 26		59 Co Cobalt 27		59 Ni Nickel 28		63.5 Cu Copper 29		65 Zn Zinc 30		70 Ga Gallium 31		73 Ge Germanium 32		75 As Arsenic 33		79 Se Selenium 34		80 Br Bromine 35		84 Kr Krypton 36	
86 Rb Rubidium 37			88 Sr Strontium 38			89 Y Yttrium 39		91 Zr Zirconium 40		93 Nb Niobium 41		96 Mo Molybdenum 42		99 Tc Technetium 43		101 Ru Ruthenium 44		103 Rh Rhodium 45		106 Pd Palladium 46		108 Ag Silver 47		112 Cd Cadmium 48		115 In Indium 49		119 Sn Tin 50		122 Sb Antimony 51		128 Te Tellurium 52		127 I Iodine 53		131 Xe Xenon 54	
133 Cs Caesium 55			137 Ba Barium 56			139 La Lanthanum 57		179 Hf Hafnium 72		181 Ta Tantalum 73		184 W Tungsten 74		186 Re Rhenium 75		190 Os Osmium 76		192 Ir Iridium 77		195 Pt Platinum 78		197 Au Gold 79		201 Hg Mercury 80		204 Tl Thallium 81		207 Pb Lead 82		209 Bi Bismuth 83		210 Po Polonium 84		210 At Astatine 85		222 Rn Radon 86	
223 Fr Francium 87			226 Ra Radium 88			227 Ac Actinium 89																															

Key

Key

A_r

Symbol

Name

Z

relative atomic mass

atomic number